Finkraft B2B E-Invoicing Platform: Architecture Design Note

Multi-ASP Dispatch, Resilient State Machine & UAE FTA Compliance

Candidate: CTO Finalist | Evaluation: Part 1 (Max 2 Pages) | Standard: Peppol PINT AE UBL 2.1

1. Business Context & Explicit Assumptions

Under the UAE Federal Tax Authority (FTA) mandate, B2B e-invoicing is governed by a 5-Corner DCTCE Model: C1 (Seller / Finkraft Customer) -> C2 (Seller ASP / Finkraft Integration Point) -> C3 (Buyer ASP) -> C4 (Buyer), with concurrent C5 (FTA) clearance. Invoice records represent sovereign tax data: zero invoices may be lost, and zero duplicates may be filed.

Explicit Architectural Assumptions:

* Compound Key Uniqueness: (seller_trn, invoice_number) is globally immutable; zero duplicate filings permitted.
* ASP A Profile: Sync POST /invoices (200), polling GET, 409 duplicate check, 10 req/s limit; times out post-acceptance.
* ASP B Profile: Async POST /v2/submissions (202), webhooks only, ZERO duplicate checks, out-of-order/duplicate webhooks.
* ASP Switching: In-flight invoices finish tracking on originating ASP; new submissions route to updated preferred ASP.

2. Unified Adapter Contract & Dynamic ASP Switching

To shield business workflows from ASP heterogeneity, Finkraft implements an extensible Adapter Pattern coordinated via ASPRegistry. Customers switch or add ASPs dynamically without codebase changes.

Method	Input Signature	Output Return	Operational / Business Responsibility
`submit_invoice`	CanonicalInvoice (PINT AE)	**SubmissionResult**	Dispatches payload; normalizes sync (ASP A) vs async job (ASP B).
`poll_status`	external_id: str	**StatusUpdate**	Polls status for GET ASPs (A); raises NotSupported for webhook-only ASPs.
`parse_webhook`	headers: dict, payload: dict	**StatusUpdate**	Verifies HMAC signature, deduplicates event ID, normalizes C3/C5 status.
`get_capabilities`	None	**ASPCapabilities**	Exposes rate limits (10 rps), sync/async traits, and native idempotency flags.

Standardized Domain Models (Summary):

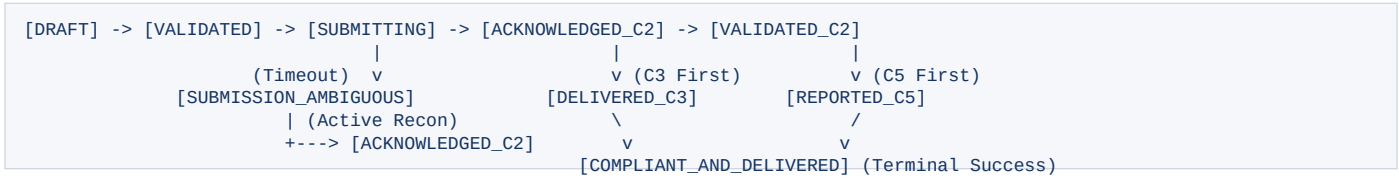
* CanonicalInvoice: Standardized payload containing seller/buyer 15-digit TRNs, issue date, amounts, and enriched PINT AE XML.
* SubmissionResult: Encapsulates success, is_ambiguous (timeouts), external_id (submission/job ID), and normalized errors.
* StatusUpdate & NormalizedError: Normalizes vendor codes (e.g. ASP A 'E104 / BT-48' vs ASP B rules) into uniform structures.

3. Invoice State Machine & Status Normalization Matrix

Finkraft enforces an internal monotonic state machine that decouples enterprise audit logs from ASP-specific statuses. State progression guarantees that both C3 Buyer Delivery and C5 FTA Clearance must be satisfied for full compliance.

ASP A Status	ASP B Status	Normalized Status	Finkraft Internal State & Milestone Resolution
`RECEIVED`	`QUEUED / ACCEPTED`	**ACKNOWLEDGED**	ACKNOWLEDGED_C2 (Payload accepted by C2 ASP)
`VALIDATED`	`N/A (Implicit)`	**VALIDATED**	VALIDATED_C2 (ASP schema & PEPPOL signature verified)
`DELIVERED (C3)`	`BUYER_RECEIVED (C3)`	**DELIVERED_C3**	DELIVERED_C3 (or COMPLIANT_AND_DELIVERED if C5 already recorded)
`REPORTED (C5)`	`TAX_CONFIRMED (C5)`	**REPORTED_C5**	REPORTED_C5 (or COMPLIANT_AND_DELIVERED if C3 already recorded)
`REJECTED`	`FAILED`	**REJECTED**	REJECTED (Terminal failure; captures BT-xx schema/tax errors)

Lifecycle State Machine Flow & Milestone Lattice



4. Failure Handling & Reliability Engineering

Submission Timeout (ASP A): *Root Cause: ASP A times out AFTER invoice accepted.*
Mitigation: State Isolation to SUBMISSION_AMBIGUOUS. Automatic blind retries strictly banned. Background active reconciliation worker queries ASP A by invoice number. If found, advances to ACKNOWLEDGED_C2; if

Duplicate Submissions (ASP B): *Root Cause: ASP B has ZERO duplicate checks. Re-submitting creates duplicate tax records at FTA.*
Mitigation: Client-Side Pessimistic Idempotency Lock: DB unique index on (seller_trn, invoice_number) + atomic compare-and-swap (VALIDATED -> SUBMITTING). In-memory repository catches DuplicateInvoiceError

Out-of-Order Webhooks (ASP B): *Root Cause: TAX_CONFIRMED (C5) arrives before BUYER_RECEIVED (C3).*
Mitigation: Milestone Lattice Resolution: Tracks independent c3_delivered_at and c5_reported_at timestamps. Invoice transitions to COMPLIANT_AND_DELIVERED as soon as both milestones are satisfied,

Late / Duplicate Webhooks: *Root Cause: ASP B re-delivers callbacks hours later.*
Mitigation: Idempotent Webhook Event Ledger: Unique constraint on (asp_id, event_id). Duplicates return HTTP 200 immediately. Monotonic transition guards reject retroactive state regressions as safe no-ops.

ASP A Rate Limiting (10 req/s): *Root Cause: Bulk upload bursts exceed ASP A quota (429).*
Mitigation: Token-Bucket Egress Throttler: Dispatcher meters traffic to 10 req/s per tenant. Excess jobs remain buffered in persistent queues with exponential jittered backoff.

5. Architectural Trade-offs & Strategic Decisions

Decision 1: Asynchronous Unified Event-Driven Pipeline vs. Synchronous Passthrough

* Strategic Choice: Unified async ingestion for both ASPs. Bulk uploads receive immediate HTTP 202 with internal receipt.
* What We Gained: Complete vendor abstraction (customers switch ASPs without API changes); absorbs ASP A 10 req/s quota; isolates transient ASP downtime inside retry queues without failing customer upload batches.

** What We Sacrificed: Single-invoice POS clients cannot receive immediate synchronous C2 clearance in the HTTP response.*

Decision 2: Pessimistic Idempotency & Reconciliation Quarantine vs. Optimistic Fast Retries

* Strategic Choice: Hard unique index on (seller_trn, invoice_number) + quarantine state (SUBMISSION_AMBIGUOUS).
* What We Gained: 100% guarantee against duplicate tax reporting to UAE FTA (zero customer tax audit penalties); deterministic audit trail for sovereign tax authorities.

** What We Sacrificed: Recovery latency (ambiguous network glitches resolve via 30s recon cycle rather than 1s instant retry).*

Finkraft 30-Day CTO Execution Plan & Launch Roadmap

Targeted Delivery for UAE FTA Mandate | Team: 3 Backend + 1 Frontend Engineers

Execution Horizon: 6 Weeks to Production Release | Deliverable: Part 3 (Strict 1-Page Budget)

1. Executive Strategy & Squad Allocation

With 3 Backend Engineers (BE1, BE2, BE3) and 1 Frontend Engineer (FE1), we operate as a single-threaded squad focused strictly on the compliant critical path. Weeks 1-4 execute feature delivery and achieve full feature freeze; Weeks 5-6 are strictly reserved for partner UAT, security penetration testing, and zero-defect launch hardening.

2. Weekly Execution Schedule (Weeks 1 to 4)

Week 1: Architectural Alignment, Scope Freeze & Core Foundations
- CTO: Enforce scope freeze with CEO/Product; audit ASP sandboxes and rate limits; establish trunk CI/CD with ephemeral previews.
- BE1 (State Machine & Locks): Core state machine, PostgreSQL schema, unique index on (seller_trn, invoice_number) with row locks.
- BE2 (ASP Adapter): Unified ASPAdapter contract for primary launch ASP (A/B), isolating authentication and rate-limiting queues.
- BE3 (PINT AE Validation): PINT AE business validation engine (15-digit TRN validation, BT-48 VAT category codes, tax arithmetic).
- FE1 (Ingestion UI): Authenticated drag-and-drop CSV/Excel bulk upload interface with instant client-side format checks.
- Milestone: Test suite passing with 100% coverage on state transitions, validation errors, and duplicate rejection.

Week 2: The 'Walking Skeleton' (End-to-End Ingestion to Clearance)
- CTO: Drive end-to-end integration across components; establish structured JSON logging with correlation IDs (seller_trn:invoice_number).
- BE1 + BE2: Wire Ingestion -> PINT AE -> ASP Adapter. Implement async worker queue and webhook ingress with HMAC verification.
- BE3: Build Active Reconciliation Worker for submission timeouts and the idempotent webhook event ledger.
- FE1: Invoice Status Dashboard (search by invoice number, state badges: ACKNOWLEDGED, DELIVERED_C3, REPORTED_C5, REJECTED).
- Milestone: First real test invoice uploaded via UI, submitted to ASP sandbox, and tracked to C5 clearance in staging.

Weeks 3-4: Failure Resilience, Bulk Scaling & Feature Freeze
- BE1 (Bulk Processing): Streaming CSV parser (10k+ rows) without memory spikes; tune token-bucket throttler (10 req/s for ASP A).
- BE2 (Secondary ASP & Failover): Finalize secondary adapter in registry to enable customer ASP switching; build Dead Letter Queues.
- BE3 (Observability & Audit): Implement tamper-evident audit logging for FTA inspection; instrument Prometheus metrics and PagerDuty.
- FE1 (Error Triage UI): Row-level error remediation view (download invalid rows with BT-xx error hints so users fix and re-upload).
- CTO + Squad: Chaos Testing Game Day (inject ASP 504 timeouts, duplicate uploads, out-of-order webhooks). FEATURE FREEZE END W4.

3. Deliberate Scope Cuts (What We Cut and Why)

* Native ERP Connectors (SAP, Oracle, Dynamics): Cut due to multi-month integration cycles. Replaced by standardized CSV/Excel uploads and REST API.
* Active-Active Multi-ASP Dynamic Load Balancing: Cut to eliminate distributed concurrency races and double-filing risks. Replaced by single ASP per seller with config switching.
* In-Browser Invoice Editor & Line-Item Correction: Cut because interactive UI tax calculators would consume 100% of FE1's runway. Replaced by 'Reject & Download Error CSV'
* Buyer Portal & Advanced Financial Analytics: Cut because compliance is driven by Seller (C1) and FTA (C5). Buyer portals add zero regulatory value for Day 1.

4. The First Two Things to Measure

Metric 1: Zero-Duplicate Submission Rate & Idempotency Violations (Target: Exactly 0)
* Rationale: Sovereign tax invariant. Submitting twice creates phantom tax liability on customer balance sheets and triggers FTA audits.
* Instrumentation: Counter finkraft_duplicates_blocked_total at edge; immediate P0 alert if any duplicate payload reaches an ASP.

Metric 2: End-to-End Compliance Latency (P95 < 120s) & Ambiguous State Dwell Time (< 60s)
* Rationale: Invoices stuck in intermediate or ambiguous states delay commercial payments. Verifies reconciliation auto-heals glitches.

Appendix: AI Usage Disclosure

Transparent Reporting of AI Tools, Methodologies, and Editorial Corrections

Candidate: CTO Finalist | Submission Requirement: Max 0.5 Page

1. AI Tools Utilized

* Antigravity / Gemini 3.8 Flash: Architecture reasoning, state machine lattice design, test suite scaffolding, and documentation.
* Google Search API: Grounding against official UAE Federal Tax Authority (FTA) Peppol PINT AE UBL 2.1 specifications.

2. What AI Was Used For

* Domain Synthesis: Validating the 5-Corner DCTCE model (C1 -> C2 -> C3 -> C4 + C5) and standard business terms (BT-31, BT-48).
* Edge-Case Modeling: Analyzing failure modes between synchronous timeout post-acceptance (ASP A) and asynchronous zero-idempotency (ASP B).
* Test Scaffolding: Accelerating implementation of 24 unit and integration tests with zero external third-party dependencies.

3. Key AI Outputs Changed or Rejected (and Why)

Rejection 1: In-Line Python Code in Executive Design Note

** AI Initial Proposal: The AI originally embedded extensive Python dataclasses directly in Section 2 of the 2-page Design Note.*

* Human Correction & Rationale: Why Rejected & Changed: The Design Note is evaluated on architectural judgment and business risk management. Raw code syntax consumed excessive vertical space and was overly developer-centric. We decoupled the Python code into docs/TECHNICAL_CONTRACT_SPEC.md and replaced it with structured functional capability tables, leaving space to elaborate on the Submission

Rejection 2: Pre-Tagged 'Scenario' Columns in Batch Ingestion CSV

** AI Initial Proposal: The initial CSV processor script relied on an artificial 'scenario' column in the CSV (e.g. TIMEOUT_RECONCILE, SCHEMA_FAIL) to branch logic.*

* Human Correction & Rationale: Why Rejected & Changed: Real enterprise ERP invoice feeds never arrive tagged with test flags; compliance engines must evaluate payloads dynamically. We stripped the scenario column completely to create a realistic business CSV, and updated the runner to automatically detect business errors (bad TRNs, duplicate keys, multi-tenant reuse) while injecting transient network failures probabilistically.

Evaluation Integrity Statement:

All architectural trade-offs, state machine transition invariants, scope prioritization decisions, and failure recovery protocols represent human CTO-level engineering leadership. AI was strictly leveraged as an acceleration tool.